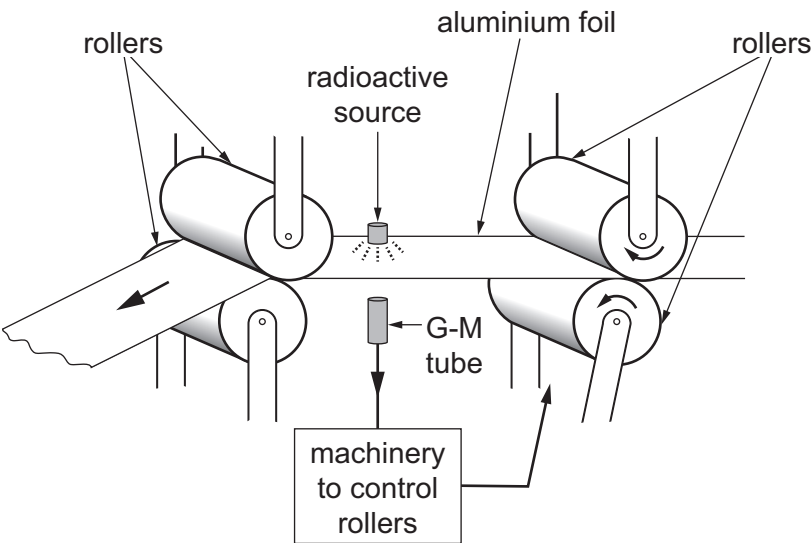


Answer all questions.

1. A group of students study the uses of radioactivity. They find that radioactive isotopes are widely used in a variety of applications. For example, alpha emitters are used in smoke detectors. Medical diagnosis and cancer treatments use a range of radioisotopes emitting alpha, beta and gamma radiation.

One use they study in detail is monitoring the thickness of aluminium foil when it is manufactured. Radiation passes through the aluminium foil and is detected by a G-M tube. Changes to the thickness cause a difference in the count rate detected and adjustments can then be made to the pressure applied by the rollers.



Different radioisotopes have different half-lives and decay in different ways. The properties of some radioisotopes are given in the table below.

Isotope	Symbol	Half-life	Decay mode
strontium-90	$^{90}_{38}\text{Sr}$	29 years	beta
americium-241	$^{241}_{95}\text{Am}$	432 years	alpha
caesium-137	$^{137}_{55}\text{Cs}$	30 years	gamma
phosphorous-32	$^{32}_{15}\text{P}$	14 days	beta
actinium-225	$^{225}_{89}\text{Ac}$	10 days	alpha



(a) Tick (✓) the boxes alongside the **three** correct statements below. [3]

- Alpha radiation consists of helium nuclei ☐
- Alpha radiation is more ionising than gamma ☐
- Strontium-90 has 90 nucleons ☐
- Gamma radiation consists of low energy waves ☐
- Gamma radiation only travels a short range in air ☐
- Beta radiation consists of slow moving electrons ☐

(b) One of the students suggests that strontium-90 is the most suitable isotope from the table for monitoring the thickness of aluminium foil. Explain whether or not you agree with this suggestion. [3]

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(c) (i) Explain what is meant by the statement: ‘The half-life of strontium-90 is 29 years.’ [2]

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(ii) Calculate the time taken for the activity of strontium-90 to fall to $\frac{1}{8}$ th of its initial value. [3]

Time = years



Examiner
only

(d) The students' teacher demonstrates experiments with radioactive sources. First she measures the radiation in the laboratory, recording 150 counts in 5 minutes. This allows her to work out the count rate of the background radiation.

(i) Calculate the background radiation count rate in counts per **second** (cps). [2]

Background count rate = cps

(ii) Suggest **two** ways in which the teacher could improve the accuracy of her result. [2]

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